

Multimodal Haptics

- Delazio, A., Israr, A., & Klatzky, R. L. (2017, June). Cross-modal correspondence between vibrations and colors. In *2017 IEEE World Haptics Conference (WHC)* (pp. 219-224). IEEE. [\[Link\]](#)

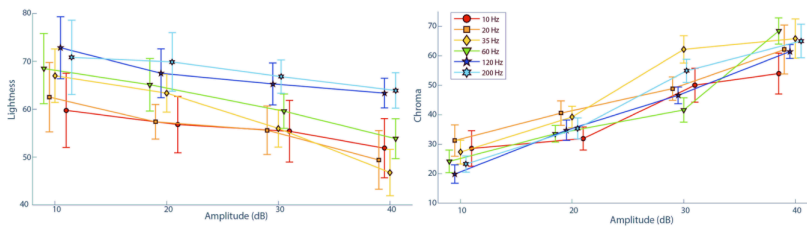


Fig. 3 Lightness and Chroma plotted as a function of stimulus amplitude (dB) at all test frequencies (Hz). Responses occur at exactly 10, 20, 30 and 40 Hz, but have been spread out for error bar clarity.

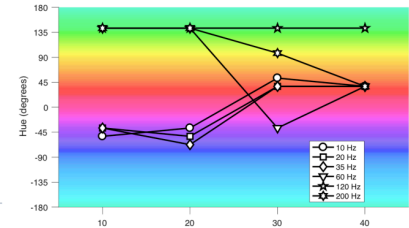


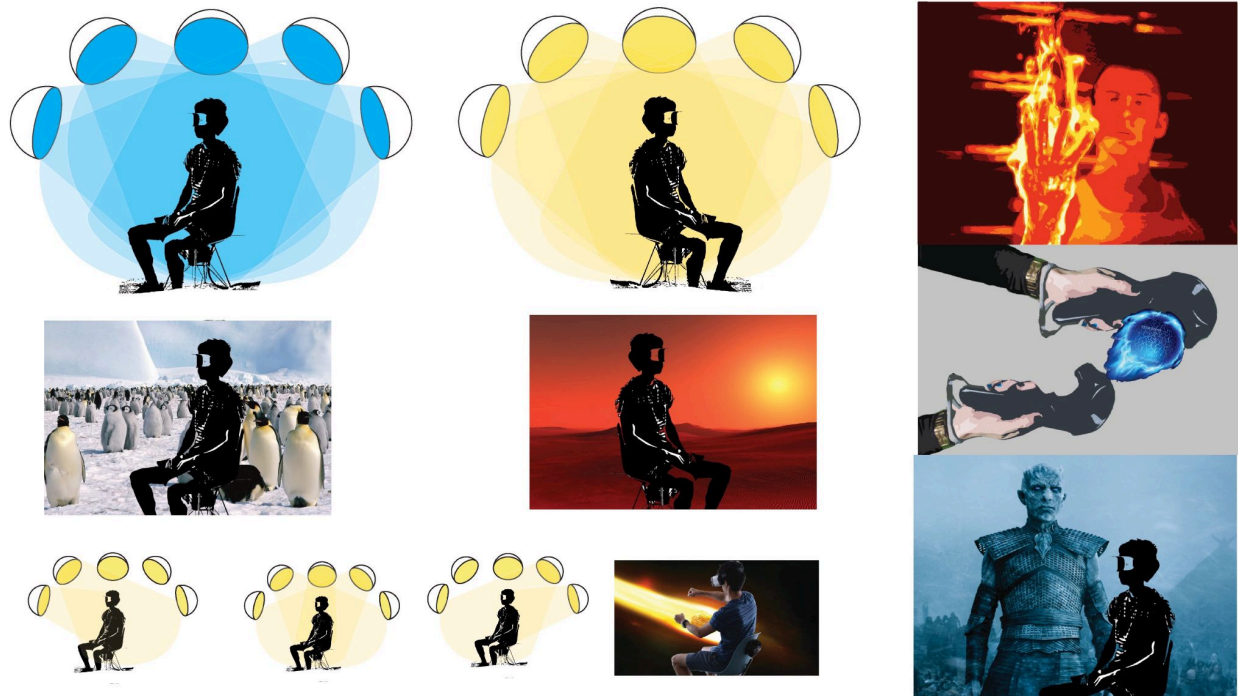
Fig 5. Trends of hue mode as a function of test frequency and amplitude.

- Ioffreda, Chris, et al. "Creating tactile content with sound." U.S. Patent No. 9,147,328. 29 Sep. 2015.

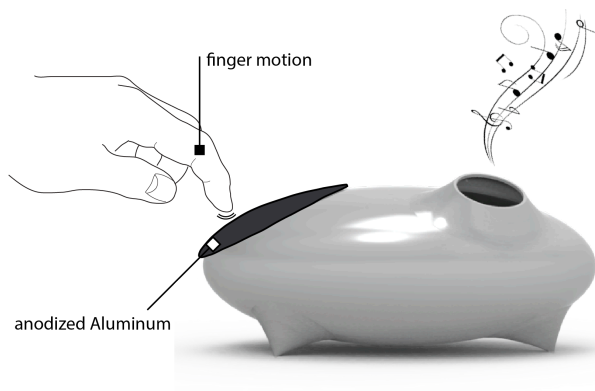
[\[Link\]](#)

- Obrist, M., Velasco, C., Vi, C. T., Ranasinghe, N., Israr, A., Cheok, A. D., ... & Gopalakrishnakone, P. (2016, May). Touch, taste, & smell user interfaces: The future of multisensory HCI. In *Proceedings of the 2016 CHI conference extended abstracts on human factors in computing systems* (pp. 3285-3292). [\[Link\]](#)

- Obrist, M., Velasco, C., Vi, C., Ranasinghe, N., Israr, A., Cheok, A., ... & Gopalakrishnakone, P. (2016). Sensing the future of HCI: Touch, taste, and smell user interfaces. *interactions*, 23(5), 40-49. [\[Link\]](#)



- Sonic Field music instrument: Israr, A., Poupyrev, I., Ishiguro, Y., Bau, O., Suzuki, Y., & Auger, J. H. (2018). *U.S. Patent No. 9,913,058*. Washington, DC: U.S. Patent and Trademark Office. [[Link](#)]



@<https://www.youtube.com/watch?v=KCRt0F1eu6U>